**Fall, 2025**

# Deep Learning and Applications: HW-1

Due date: 2025/10/17 23:59

## Instructions:

1. Please complete the tasks using Python, and submit the code in an executable `.ipynb` (Jupyter Notebook) file. The notebook must run from top to bottom without manual intervention. Upload the final version to EEClass.

2. The use of automatic differentiation tools (e.g., TensorFlow, PyTorch, Keras, MXNet) is strictly prohibited.The backpropagation algorithm must be implemented manually using NumPy for all numerical computations. Other libraries may be used only for plotting, or result visualization.

## Question1:

Consider a feedforward neural network with two hidden layers for a three-class classification task. The input is two-dimensional ($x \in \mathbb{R}^2$). Each hidden layer contains two neurons with sigmoid activations, and the output layer uses a softmax activation. Train the network with gradient descent using a learning rate of $\eta = 0.05$ for 5 iterations; each iteration uses the entire batch (all three samples). Use the given initial parameters $(W_1, b_1, W_2, b_2, W_3, b_3)$ and the given dataset $\mathcal{D}$:

$$W_1 = \begin{bmatrix} 0.10 & -0.20 \\ 0.30 & 0.05 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0.00 \\ 0.00 \end{bmatrix},$$

$$W_2 = \begin{bmatrix} 0.02 & -0.04 \\ -0.03 & 0.06 \end{bmatrix}, \quad b_2 = \begin{bmatrix} 0.01 \\ -0.02 \end{bmatrix},$$

$$W_3 = \begin{bmatrix} 0.08 & -0.05 & 0.02 \\ -0.04 & 0.07 & 0.03 \end{bmatrix}^T, \quad b_3 = \begin{bmatrix} 0.00 \\ 0.00 \\ 0.00 \end{bmatrix}.$$

$$\mathcal{D} = \left\{ \left( \begin{bmatrix} -1.0 \\ 1.5 \end{bmatrix}, 0 \right), \left( \begin{bmatrix} 1.0 \\ -1.0 \end{bmatrix}, 1 \right), \left( \begin{bmatrix} 2.0 \\ 0.5 \end{bmatrix}, 2 \right) \right\}.$$

From the given initialization, run **five** full-batch iterations (batch = all 3 samples; one iteration = one epoch). At each iteration, record the loss and the updated parameters $\Theta(W, b)$. The loss is the *average cross-entropy* over the batch:

$$\mathcal{L}(\Theta) = \frac{1}{N} \sum_{n=1}^{N} E_n.$$

## Question2:

Implement a Deep Neural Network (DNN) model to classify handwritten digits using the MNIST dataset. The MNIST dataset consists of grayscale images of handwritten digits ranging from 0 to 9. The You are required to download the dataset using PyTorch, and

PyTorch should only be used for this purpose. Each sample is a $28 \times 28$ grayscale image representing a single digit.
The dataset contains 60,000 training samples and 10,000 testing samples. To properly evaluate model generalization, the training data will be further divided into 50,000 training and 10,000 validation samples. Since a DNN expects a 1D input vector, please flatten each image before feeding it into the network.

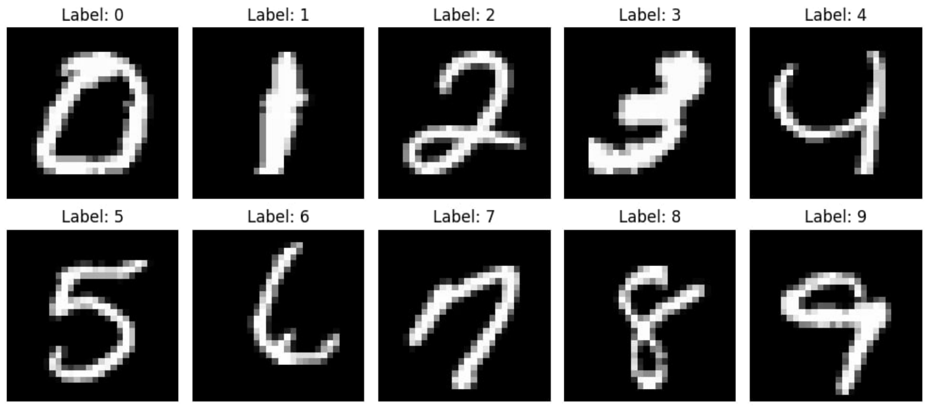


Figure 1: Sample images from the dataset.

(a) Please construct a Deep Neural Network (DNN) for multi-class classification. For $N$ samples and $K$ categories, the cross-entropy objective function is defined as:

$$E(\mathbf{w}) = -\sum_{n=1}^{N}\sum_{k=1}^{K} t_{nk} \log y_k(\mathbf{x}_n, \mathbf{w}).$$

Then, minimize $E(\mathbf{w})$ by running the error backpropagation algorithm using Stochastic Gradient Descent (SGD):

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \nabla E(\mathbf{w}^{(t)}).$$

The following hyperparameters are to be determined: number of hidden layers, number of hidden units, learning rate, number of epochs and mini-batch size. The report shall include (a) the learning curves for both training and validation, (b) the accuracy for both training and validation, and (c) the test error rate.

(b) Perform both **zero initialization** and **random initialization** for the model weights. Compare the corresponding error rates and learning behaviors, and discuss the differences between the two initialization schemes.

(c) Compare the model performance with and without the inclusion of the bias term $b$. Discuss the effect of bias initialization in the report.